

AURORA-12

A fully synthetic benchmark for evidence-linked poster generation

PostEx Synthetic Research Collective

Fictional Methods Laboratory · Example Research Campus · no real authors or institutions

POSTEX-SYNTH-2026-001 · fully synthetic · CC0-1.0 · no DOI

Fictional lab logo

1,240
synthetic
profiles

12
anonymous
model features

4
deterministic
cohort
simulations

Communication test

Can a realistic multi-cohort manuscript exercise evidence-linked poster generation without redistributing a real paper or implying biomedical evidence?

Evidence: Abstract and Introduction, pp. 1-2

Deterministic workflow

- 1
- Fixed seed generated 1,240 profiles
- 2
- 480 anonymous transcript-like features
- 3
- Nested elastic-net selection
- 4
- 12-feature score and shift testing

FOUR SYNTHETIC COHORTS

Four deterministic synthetic cohorts

• Discovery n=620 · Internal validation n=310

• External A n=186 · External B n=124

• No participants, specimens or external datasets

Evidence: Abstract and Methods, pp. 1-2

Generation to validation

Synthetic benchmark workflow

All profiles, features and outcomes are authored simulation fixtures

+

Fixed seed

1,240 profiles
480 features

Seed 20260808

Latent modules

3 anonymous axes
controlled censoring

No gene mapping

Elastic net

480 to 12 features
nested resampling

AURORA-12

D I

A B

Four cohorts

shift and stress tests

Interpretation boundary

Methodological fixture only - no biological or clinical inference

Figure 1 · Deterministic generation and validation workflow

Authored synthetic fixture · manuscript p. 2

External C-indices remained 0.75 and 0.74

Synthetic baseline: 0.66 and 0.65 · software fixture, not clinical validation

Evidence: Results, pp. 1 and 3; Figure 2

Discrimination across shifts

Authored discrimination benchmark

Concordance index - synthetic values only

Cohort	AURORA-12	Five-feature baseline
Discovery	0.78	0.68
Internal	0.76	0.67
External A	0.75	0.66
External B	0.74	0.65

Figure 2 · Authored discrimination values across cohort shifts

Authored synthetic fixture · manuscript p. 3

Calibration and stress boundary

- External calibration slopes were 0.97 and 0.95.
- External absolute errors were 0.031 and 0.038.
- Aggregate coefficient-sign stability was 92% across 500 resamples.
- The most severe pre-specified shift reduced C-index by 0.04.

Evidence: Results, pp. 1, 3-4; Figures 3-4

External validation retained discrimination: C-index 0.75 / 0.74

Calibration & stability

92%
coefficient-sign stability

0.97 / 0.95
external calibration slopes

0.031 / 0.038
external absolute errors

Evidence: Stability and calibration, p. 4

Calibration and coefficient stability

Authored evaluation values - no patient-level data

24-month calibration

500 bootstrap resamples

92%
sign stability

Calibration slope
0.95-0.99

Absolute error
0.021-0.038

Stable features
11 / 12

Synthetic stability supports fixture QA only

Figure 3 · Calibration and coefficient-sign stability

Authored synthetic fixture · manuscript p. 4

Pre-specified stress test

Pre-specified stress-test boundary

Increasing batch shift and latent-module imbalance

Shift Level	C-index
Reference	0.78
Mild	0.76
Moderate	0.74
High	0.72
Severe	0.70

Maximum C-index drop: 0.04

Do not interpret this authored curve as biological transportability or clinical robustness

Figure 4 · Stress-test boundary under increasing shift

Authored synthetic fixture · manuscript p. 4

Supported conclusion

A rights-safe methodological test fixture

- No biological, prognostic or clinical inference
- All cohorts, values and identities are fictional

Evidence: Discussion and Conclusion, p. 5

Original synthetic fixture · CC0-1.0 · evidence coverage 100% · local-only processing

Approvals current · synthetic boundary locked · editable logo placeholder